Example 1: Write a Java program to store and display the maximum daily temperatures in a week.

```java
import java.util.Scanner;
public class TemperatureReadings {
     public static void main(String[] args) {
     Scanner keyboard = new Scanner(System.in);
     double[] temperature; // Create the array
     temperature = new double[7]; // Allocate a 7 elements
     // Enter temperatures
     for (int i = 0; i < temperature.length; i++)
     {
     System.out.println("enter max temperature for day " + (i+1));
     temperature[i] = keyboard.nextDouble();
     }
     keyboard.close();
     // Display Temperatures
     System.out.println(); // blank line
     System.out.println("***TEMPERATURES ENTERED***");
     System.out.println(); // blank line
     for (int i = 0; i < temperature.length; i++)
          System.out.println("day "+(i+1)+" "+ temperature[i]);
     }
}
```

```
Problems  Javadoc  Declaration  Console  Progress
<terminated> TemperatureReadings [Java Application] /Library/Java/JavaVirtualM
enter max temperature for day 1
23
enter max temperature for day 2
25
enter max temperature for day 3
22
enter max temperature for day 4
26
enter max temperature for day 5
27
enter max temperature for day 6
25
enter max temperature for day 7
30

***TEMPERATURES ENTERED***

day 1 23.0
day 2 25.0
day 3 22.0
day 4 26.0
day 5 27.0
day 6 25.0
day 7 30.0
```

Example 2: Write a Java program to store a week temperature and print the highest temperature.

```java
import java.util.Scanner;
public class HighestTemperature {
	public static void main(String[] args) {
	Scanner keyboard = new Scanner(System.in);
	double[] temperature; // Create the array
	temperature = new double[7]; // Allocate a 7 items in the array
	// Enter Temperatures
	for (int i = 0; i < temperature.length; i++)
	{
	System.out.println("enter max temperature for day " + (i+1));
	temperature[i] = keyboard.nextDouble();
	}
	keyboard.close();

	double highest=temperature[0];// Assume the first item is the highest
	//This loop runs from the 2nd item to the last item in the array
	for (int i = 1; i < temperature.length; i++)
		if (temperature[i] > highest)
			highest=temperature[i]; // Reset result to new highest
	System.out.println("The highest temperature of the week " + highest);
	}
}
```

```
Problems  Javadoc  Declaration  Console  Progress
<terminated> HighestTemperature [Java Application] /Library/Java/JavaVirtualM
enter max temperature for day 1
25
enter max temperature for day 2
27
enter max temperature for day 3
26
enter max temperature for day 4
30
enter max temperature for day 5
29
enter max temperature for day 6
28
enter max temperature for day 7
24
The highest temperature of the week 30.0
```

Example 3: Write a Java program to print even numbers in an array.

```
public class ArrayEvenNumber {
	public static void main(String[] args) {
	int numbers[] = {5, 2, 6, 3, 4};
	for (int i=0; i < numbers.length; i++)
		if (numbers[i] %2 == 0)
			System.out.println(numbers[i]);
	}
}
```

```
Problems  @ Javadoc  Declaration  Console  Progress
<terminated> ArrayEvenNumber [Java Application] /Library/Java/JavaVirtualMac
2
6
4
```

Example 4: Write a Java program to print the average of numbers in an array.

```
public class ArrayAverage {
	public static void main(String[] args) {
	double[] array= {22.4, 13.3, 9.5, 6, 10.3};
	double sum=0;
	for (int c = 0; c < array.length; c++)
		sum=sum+array[c];
	System.out.println("The average = " + sum/array.length);
	}
}
```

```
Problems  @ Javadoc  Declaration  Console  Progress
<terminated> ArrayAverage [Java Application] /Library/Java/JavaVirtualMachines
The average = 12.3
```

Example 5: Write a Java program to search an element in an array.

```
import java.util.Scanner;
public class ArraySearch {
	public static void main(String[] args) {
	int numbers[] = {5, 2, 7, 4, 10};
	int i,num,flag=0;
	Scanner s = new Scanner(System.in);
	System.out.print("Enter the element you search for : ");
	num = s.nextInt();
	s.close();

	for (i=0; i < numbers.length; i++)
		if (num==numbers[i])
		{
			flag=1;
			break;
		}
```

```
        if (flag==1)
            System.out.println("The Element Found at position: "+ i);
        else
            System.out.println("The Element Not Found");
        }
}
```

```
Problems  Javadoc  Declaration  Console  Progress
<terminated> ArraySearch [Java Application] /Library/Java/JavaVirtualMachines/
Enter the element you search for : 7
The Element Found at position: 2
```

```
Problems  Javadoc  Declaration  Console  Progress
<terminated> ArraySearch [Java Application] /Library/Java/JavaVirtualMachines/
Enter the element you search for : 3
The Element Not Found
```

Example 6: Write a Java program to sort the items of an array in ascending order.

```
public class ArrayAscendingOrder {
        public static void main(String[] args) {
            double[] array= {22.4, 13.3, 9.5, 6, 10.3};
            double temp;

          for (int i = 0; i < array.length; i++)
          {
              for (int j = i + 1; j < array.length; j++)
              {
                  if (array[i] > array[j])
                  {
                      temp = array[i];
                      array[i] = array[j];
                      array[j] = temp;
                  }
              }
          }
          System.out.print("Ascending Order: ");
          for (int i = 0; i < array.length; i++)
            System.out.print(array[i] + " ");
      }
}
```

```
Problems  Javadoc  Declaration  Console  Progress
<terminated> ArrayAscendingOrder [Java Application] /Library/Java/JavaVirtualM
Ascending Order: 6.0 9.5 10.3 13.3 22.4
```

Homework:

Q> Write a Java program to count the number of occurrences of an element in an array.

Q> Write a Java program to print the prime numbers in an array.

Q> Write a Java program to sort the items of an array in descending order.

Q> Write a Java program to sort names of an array in an alphabetical order.